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Reg. No: .....

Name: .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION October 2026

FIFTH SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCZGY300, MG5DSCZIM300 -

ANIMAL DIVERSITY CHORDATA II

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create(C), Skill(S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

PART A

Multiple Choice Questions

Answer **ALL** questions.

Each question carries **1** mark.

1. Which of the following superorders includes flightless birds such as *Struthio*? (K)/(CO 1)
   (a) Neognathae (b) Paleognathae (c) Archaeornithes (d) Odontognathae

2. In *Columba livia*, the thin-walled, dilated part of the oesophagus used for temporary food storage is the: (K)/(CO 2)
   (a) Gizzard (b) Proventriculus (c) Crop (d) Caecum

3. The Malabar Grey Hornbill is an endemic bird species associated with which region? (K)/(CO 3)
   (a) Eastern Ghats (b) Western Ghats (c) Himalayas (d) Sundarbans

4. Egg-laying mammals like *Platypus* belong to which order? (K)/(CO 1)
   (a) Tachyglossa (b) Platypoda (c) Metatheria (d) Macroscelidea

5. The process of re-ingesting soft faecal pellets to maximize nutrient absorption in rabbits is known as: (U)/(CO 2)
   (a) Primary digestion (b) Ruminant digestion (c) Coprophagy (Secondary digestion) (d) Mastication

6. Which of the following aquatic mammals belongs to the Order Sirenia? (K)/(CO 3)
   (a) *Delphinus* (b) *Dugong* (c) *Pteropus* (d) *Procavia*

7. *Archaeopteryx* serves as an evolutionary bridge between reptiles and birds and belongs to the subclass: (K)/(CO 1)

(a) Neornithes (b) Prototheria (c) Archaeornithes (d) Theria

8. To which mammalian order does the armadillo belong? (K)/(CO 1)

(a) Xenarthra (b) Pholidota (c) Tubulidentata (d) Erinaceomorpha

9. Which gland is uniquely specialized in mammals for nourishing offspring? (U)/(CO 2)

(a) Sebaceous gland (b) Mammary gland (c) Harderian gland (d) Scent gland

10. Tree shrews are categorized under which order of Class Mammalia? (K)/(CO 1)

(a) Macroscelidea (b) Soricomorpha (c) Scandentia (d) Hyracoidea

11. Pneumatic bones in birds serve primarily to aid in: (U)/(CO 3)

(a) Digestion (b) Reduction of body mass for flight (c) Excretion (d) Reproduction

12. In the comparative study of vertebrates, mesonephric kidneys are characteristic of adult: (An)/(CO 2)

(a) *Euphlyctis* (b) Pigeon (c) Rabbit (d) Both Pigeon and Rabbit

PART B

Short Answer Questions

Answer any **6** questions

Each question carries **3** marks

13. State three distinct reptilian characteristics present in *Archaeopteryx*. (U)/(CO 1)
14. Explain the dental formula and heterodont dentition of the rabbit. (U)/(CO 2)
15. Mention three structural flight adaptations found in the skeletal framework of birds. (U)/(CO 3)
16. Describe the key characteristics of Subclass Prototheria with suitable examples. (K)/(CO 1)
17. Briefly explain the physiological significance of secondary digestion in rabbits. (U)/(CO 2)
18. List three distinct mammalian orders under Eutheria along with one representative example for each. (K)/(CO 1)
19. Give a brief account on excretory system of birds using suitable type specimen. (U)/(CO 2)
20. Differentiate the structure of locomotor organs among *Euphlyctis*, Pigeon, and Rabbit. (An)/(CO 2)

PART C

Short Essay Questions

Answer any **4**questions

Each question carries **5** marks.

21. Give a classification of Subclass Neornithes up to superorders, highlighting key diagnostic features and examples. (U)/(CO 1)

22. Describe the respiratory system of *Columba livia* and explain the mechanism of double respiration. (U)/ (CO 2)

23. Mention your understanding of the term ‘comparative anatomy’. Also explain how this benefits the study of various organisms. (An)/(CO 2)

24. Describe the axial and appendicular skeleton of the rabbit, excluding the skull. (An)/(CO 2)

25. Discuss the anatomical and physiological adaptations of aquatic mammals using examples from Sirenia and Cetacea. (U)/ (CO 3)

26. Give a comparative account of the heart and aortic arches in *Euphlyctis*, Pigeon, and Rabbit. (An)/(CO 2)